Transcript for Module 3 Deep Dive on the AWS Well-Architected Tool

1.1 Welcome!

Welcome to module three of AWS Well-Architected: Deep Dive on the AWS Well-Architected Tool.

1.2 Learning objectives

In this module you will learn about the components and features of the AWS Well-Architected Tool. You will also discover how to use the tool to do a Well-Architected Framework Review and where to learn more about the tool.

1.3 A mechanism for continuous improvement

To achieve the desired goal from a framework review, it's important to consider it as a step in a continuous improvement plan that integrates with your workload lifecycle. This mechanism starts with first learning the strategies and best practices for architecting in the cloud. Next, you can measure your architecture using the framework, AWS Well-Architected lenses, and your organization's best practices with the custom lenses in the AWS Well-Architected Tool. Finally, you can use the outcome from that to improve your cloud architecture by addressing any high-risk issues. These issues can be identified using improvement plans, Well-Architected Labs, the AWS Partner Network, AWS solutions architecture teams, and more. This three-step mechanism has to be consistently applied on every workload in your organization. A workload identifies a set of components that together deliver business value. You will learn more about workload details in a later module.

1.4 Components of the Well-Architected Framework

You saw a version of this diagram of the framework components in an earlier module. To review, the framework includes content that you can use to learn AWS best practices. It also has a tool that can help measure your workload and teams against best practices. Additionally, the tool contains data that you acquire during the review of your workloads. This can be used to continuously improve your workloads and operations. In this module, you will dive deeper into how the AWS Well-Architected Tool can be used by customers to measure and improve over time.

1.5 AWS Well-Architected Tool

The AWS Well-Architected Tool is designed to help you review the state of your applications and workloads, providing a central place for architectural best practices and guidance. In addition to the standard guidance provided by the framework and AWS lenses, the tool helps you add best practice guidance using custom lenses. The quickest way to get started is to complete a Well-Architected Framework Review, using the tool in the console or with the APIs. You can create the AWS Well-Architected Tool workload in the customer AWS account to store it securely. This follows least-privilege access, with only the right people having access to workload details. Workloads can be shared with your solutions architect and account team or partner resource for collaboration on review or remediation steps. Custom lenses can also be shared with the solutions architect or partner resource for collaboration.

1.6 AWS Well-Architected Tool overview

When you navigate to the AWS Well-Architected Tool in the AWS Management Console, the dashboard shows the resources for the Region you have selected. As mentioned in the history of Well-Architected, continual updates and improvements to the framework and the tool are made based on customer feedback. If you have workloads that are not using the most recent version of a lens or the framework, you will have an option to upgrade those workloads using the View available upgrades option in the console. You can access a list of all current workloads available in a particular account and Region in the Workloads list in the console. You can also view workload details such as name, owner, number of questions answered, and number of risks identified. From the workloads console, you also have the ability to define a new workload to start a new review for that workload.

1.7 AWS Well-Architected Tool new workload

When creating a new workload in the tool, the first thing you need is a unique and descriptive name to identify it. You also need to specify a description of the workload to document its scope and intended purpose. Another mandatory field is the review owner field, which was added in 2020. The review owner is the person who ultimately has responsibility for completing the review, reporting the status of risks, and tracking improvements over time. Next is the environment for the workload. The two choices, production and pre-production, help you identify where your workload is in its lifecycle.

1.8 AWS Well-Architected Tool new workload (cont.)

The last mandatory fields, when setting up a new workload, are the Regions in which it runs. When creating a new workload in the tool, remember that the concept of workload is something you elect as a customer. That means it is a synthetic concept, so what Regions that the workload runs in are primarily used to help search, sort, and filter. Also note that the best practices in the framework can be extended beyond just your AWS environment. You can use the tool to review resources that are running on premises or in other cloud provider environments.

The next section is for account ID, or IDs, if your workload spans multiple accounts. Note that by default you will not need to give the tool any AWS Identity and Access Management (IAM) permissions or access to the account IDs that are specified here. If you are using the Well-Architected API, or services provided by AWS software partners that work across accounts, you might need to change permissions in the account IDs listed for programmatic access to work correctly. Consult the API reference in the tool documentation or your third-party or AWS Marketplace partner product documentation.

There is also an optional location to specify the URL of an architectural design for the workload you will be doing the review on. This can be used as a resource for reference when running a review to ensure that everyone has the same setup information about the workload. This is especially helpful if your review is being supported by AWS employees or an AWS Well-Architected partner.

The last set of options when creating a new workload is to establish the industry type and industry for your organization or workload. This is again provided primarily to assist you in sorting, filtering, and searching when you create multiple workloads.

1.9 AWS Well-Architected Tool workload details

After a new workload has been created, or selected from a list of existing workloads, you will find an overview of the workload. In the overview, you have the ability to edit the details of the workload after it has been created. You can also delete a workload from this view. The overview of the workload includes the number of questions that would be been answered out of the total available. It also contains the risks that have been identified during the review process. It is important to note on the workload details page that you have the option to save a milestone for tracking progress when doing a framework review.

1.10 AWS Well-Architected Tool (workload details - cont.)

Below the workload details, you will also see any notes that have been entered about the workload overall. These notes are open ended and can be used to track progress, important changes, or upcoming launch information. Below the workload notes, you will see a list of the lenses applied to this workload. Lenses outside of the default framework are covered in another session in the series. This includes both the lenses created and supported by AWS and also custom lens creation. For more information, see the other sessions. At the bottom of the workload details, you will notice the pillar priority that you have chosen for this workload. Listed here is the default pillar priority. This is not to say that any pillar is more important than another pillar. Rather, this is the order that most customers tend to approach the concepts in the framework. If your priorities for the workload vary from the default order, you can use the edit button to change the pillar priority. Changing the pillar priority will rearrange the order of the questions in the tool. It will also change the order of the recommendations for improvement based on your priorities.

1.11 AWS Well-Architected Tool (milestones)

Milestones are a mechanism that you can use to track the changes that happen to a workload over time. Typically, a milestone is created after you have completed your initial review. As you make improvements or changes to your workload, you can update your answers, notes, and best practices to denote those changes. After those updates have been made, you can create a new milestone to help track progress and improvements over time. Select any of the milestones you have created by name, and then choose Generate report. The report will be based on the status of the review at the point in time that the milestone was saved. You can also view any milestones, including the current one, using View milestones.

1.12 AWS Well-Architected Tool (sharing)

One other important feature in the tool is the ability to share workloads with other users, accounts, or even through AWS organizations. If you share a workload with multiple principals, there is an option to search or filter to help find current workload shares. Through the sharing details, you can view what the current principal is and whether it is an IAM user, another AWS account, or an AWS organization. You are also able to review the current status of a share, whether it is pending or accepted, and what permissions it grants other principals.

1.13 AWS Well-Architected Tool (content)

Now, take a moment to dive deeper and consider how a question in the tool appears. At the top, you will find the pillar and question number, including the question that is the topic of discussion. Each question has a key concept that it relates to. In this case, the Cost 6 question is focused on rightsizing. Each concept in a question in the framework is designed to help you understand how to implement the design principle for that pillar. For example, in this case, the concept of rightsizing is used to help you implement the design principle of adopting a consumption model.

Below the question and its explanation, you will also find a list of check boxes that represent the best practices for how to achieve that key concept. In the detail bar on the left of the screen, you can explore additional explanation about each best practice. You can also access helpful resources to learn more about potential methods for implementing best practices in that question.

1.14 AWS Well-Architected Tool (custom lenses)

In addition to measuring your workloads against the framework and AWS lenses, you can now create your own custom lenses. Custom lenses can include pillars, questions, answer choices, helpful resources, and improvement plans. You can specify rules to determine which options, when not followed, would result in a high risk or medium risk. Then, you can provide your own guidance to resolve the risk. This helps you share these lenses across your accounts and consistently measure workloads in your organization. You can create your own custom lens using the AWS Well-Architected Tool. Download the JSON template, and then upload your lens and apply it to your workload review just as you apply AWS lenses today. This helps customers review your workload using your customized set of questions in the same way you review workloads against the framework or AWS content. You can also share the custom lens with another AWS account, solutions architect, or partner.

1.15 What's new with AWS Well-Architected?

AWS introduced the sustainability pillar during re:Invent 2021 to help customers minimize the environmental impacts of running cloud workloads. In March 2022, this pillar became available for customers to use during workload reviews in the AWS Well-Architected Tool. The sustainability pillar is designed to help chief technology officers, architects, developers, and operations team members contribute to an increasing number of sustainability targets set by their organizations. The tool now features direct access to AWS re:Post, a community-driven, question-and-answer service to help AWS customers remove technical roadblocks, accelerate innovation, and enhance operation. AWS re:Post includes 40-plus topics, with a community specific to AWS Well-Architected.

The AWS Well-Architected Tool launched integration with AWS Organizations in June 2022. This integration helped cloud architects share workloads and custom lenses more broadly across their organization. AWS Organizations is an account management service that customers use to consolidate multiple AWS accounts into a single, centrally managed organization. This update increases efficiency and streamlines the sharing of lenses and workloads across multiple accounts. The tool launched in AWS GovCloud (US) Regions in August 2022 for customers with specific regulatory and compliance requirements and AWS Partners. Both the public and commercial sectors can use it to conduct self-service Well-Architected Reviews. AWS GovCloud (US) is an isolated Region designed to host sensitive data and regulated workloads in the cloud. With the AWS Trusted Advisor integration, the AWS Well-Architected Tool now presents findings based on automated Trusted Advisor resource checks. This provides further contextual information during your reviews, improving the accuracy of your answers and speed of your review.

Previously, customers conducting framework reviews had to spend time verifying their answers by double-checking their workloads to check if they were following best practices. There was not a clear link between the workload being reviewed and its associated resources. The AWS Well-Architected Tool is also now integrated with AWS Service Catalog AppRegistry. You can use AppRegistry to store your AWS applications, associated resource collections, and application attribute groups. The integration with AppRegistry gives you better visibility into which applications are associated with which workloads during the review process. This can save you time by tracking and organizing your workloads' associated resources.

1.20 Summary

In this module, you have learned about the AWS Well-Architected Tool. In particular, you learned about the components and features of the tool and how to use it to do a Well-Architected Framework Review. Finally, you also discovered where to go to learn more about the AWS Well-Architected Tool.

1.21 Thank you

Thank you for taking the time to learn about the AWS Well-Architected Tool. Please consider our other learning modules as you continue on your Well-Architected journey.